Rocket Science and Technology 4363 Motor Ave., Culver City, CA 90232

Phone: (310) 839-8956 Fax: (310) 839-8855

Longitudinal Body Bending Dynamics

05 August 2013

By C. P. Hoult

Introduction

Whenever nonrigid (flexible) body effects are considered such as propellant sloshing, axial vibrations or body bending Lagrange's equations come into play. In this memo, the free-free behavior of a uniform non-rolling beam is first estimated. The resulting mode shapes are then used in a Rayleigh-Ritz approximation to find the natural frequencies and mode shapes of a non-uniform beam. The results are implemented in an Excel™ code named FLEXIT.xls.

Nomenclature

Mnemonic	Definition
EI	Bending stiffness, lb-in^2
$\overline{EI}$	Average bending stiffness, lb-in^2
L	Beam length, in
μ	Mass per unit length, sl/in
$\overline{\mu}$	Average mass per unit length, sl/in
N	Number of normal modes used to approximate the beam displacement,
$q_i(t)$	Amplitude of the ith normal mode, ft
ω_n	Natural frequency of the nth mode, red/sec
$W_n(x)$	Normalized shape of the nth mode
T	Kinetic energy
U	Potential energy
t	Time, sec
x	Distance aft of the nose tip, in
y	Beam lateral displacement, in
$\eta(x)$	Normal mode shape

Uniform Beam

The solution for the bending vibrations of a uniform, non-spinning beam are based on the material in ref. (2), pp.67-80:

$$a^2 = \frac{\overline{EI}}{\overline{\mu}} \tag{1}$$

where the beam-average properties are denoted by an over bar:

$$\overline{EI} = \frac{1}{L}\int_0^L EI(x)dx$$

(2a)
and

$$\overline{\mu} = \frac{1}{L}\int_0^L \mu(x)dx$$

(2b)

The characteristic equation is

$$\cos(\sqrt{\frac{\omega_n}{a}}L)\cosh(\sqrt{\frac{\omega_n}{a}}L) = 1$$

(3)

This has roots

$$\sqrt{\frac{\omega_n}{a}}L \cong 0, \frac{5\pi}{2}, \frac{7\pi}{2}.... = \frac{(2n+1)\pi}{2}$$

(4a)

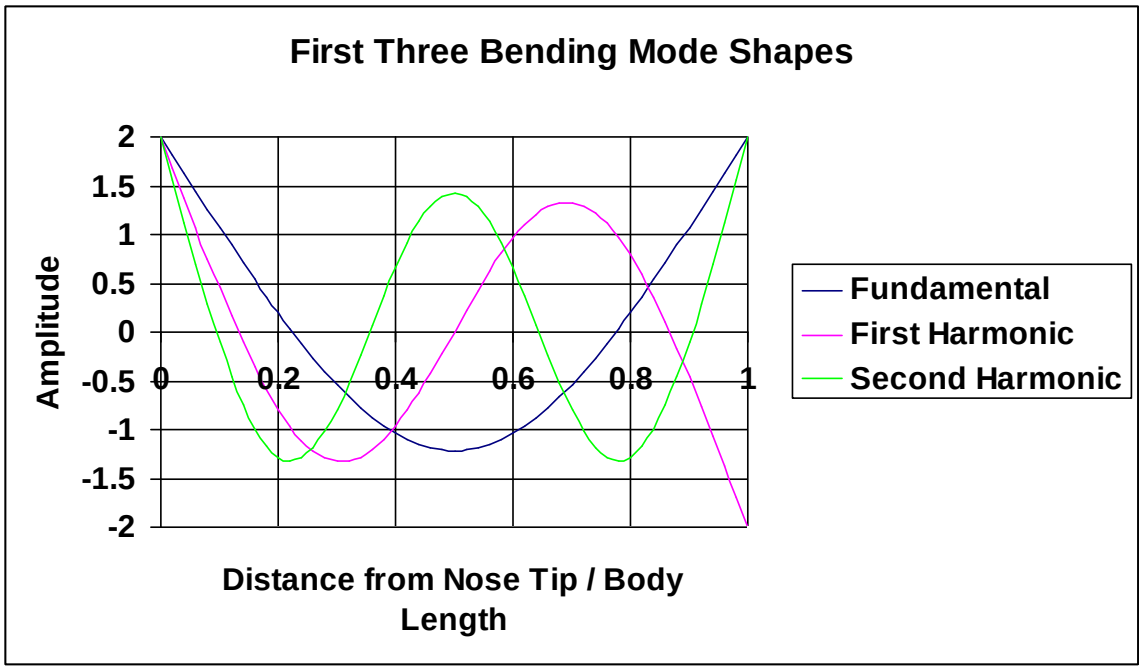
More exactly, $$\sqrt{\frac{\omega_n}{a}}L = 0, 1.50561873\pi, 2.4997527\pi, 3.5\pi....$$

(4b)

The mode shapes are

$$W_n(x) = \left[\frac{\cos\left(\sqrt{\frac{\omega_n}{a}}L\right) - \cosh\left(\sqrt{\frac{\omega_n}{a}}L\right)}{\sinh\left(\sqrt{\frac{\omega_n}{a}}L\right) - \sin\left(\sqrt{\frac{\omega_n}{a}}L\right)}\right]\left[\sinh\left(\sqrt{\frac{\omega_n}{a}}x\right) + \sin\left(\sqrt{\frac{\omega_n}{a}}x\right)\right] + \cosh\left(\sqrt{\frac{\omega_n}{a}}x\right) + \cos\left(\sqrt{\frac{\omega_n}{a}}x\right)$$

(5)



These are the first three mode shapes for a uniform, free-free, non-rolling beam as developed from eq's. (4b) and (5). The source is FLEXIT.xls which was also used to find the natural frequencies:

Mode	Natural Frequency, rad/sec
Fundamental	277.452
First Harmonic	764.806
Second Harmonic	1499.317

Lagrange's Equations

To estimate the frequencies and mode shapes of a non-uniform beam requires the solution of Lagrange's equations using the Rayleigh-Ritz method. Start by assuming the beam displacement is approximated by

$$y(x) = \sum_{i=1}^{N} q_i W_i(x).$$

(6)

Then, Lagrange's equations are

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) + \frac{\partial U}{\partial q_i} = 0 \qquad (7)$$

The system kinetic energy is

$$T \equiv \frac{1}{2}\int_0^L \mu(x)(\dot{y}(x))^2 dx$$

(8)

where the beam displacement is assumed to be represented by a sum of N mode shapes satisfying the boundary conditions...the Rayleigh-Ritz technique:

$$\dot{y}(x) = \frac{\partial y}{\partial t} = \sum_{i=1}^{N} \dot{q}_i W_i(x)$$

(9)

Next, assume simple harmonic motion so that

$$q_i = A_i \exp(j\omega t) \qquad (10)$$

and

$$\dot{y}(x) = \sum_{l=1}^{N} j\omega q_i W_i(x), \text{ and}$$

(11a)

$$\ddot{y}(x) = -\sum_{i=1}^{N} \omega^2 q_i W_i(x).$$

(11b)

With this, the kinetic energy becomes

$$T = \frac{1}{2}\int_0^L \mu(x)\left[\sum_{i=1}^{N} \dot{q}_i W_i(x)\right]^2 dx,$$

(12)

$$\frac{\partial T}{\partial \dot{q}_i} = \int_0^L \mu(x)\left[\sum_{l=1}^{N} \dot{q}_l W_l(x)\right] W_i(x) dx, \text{ or}$$

(13)

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) = \int_0^L \mu(x)\left[\sum_{1=1}^{N} \ddot{q}_l W_l(x)\right] W_i(x) dx.$$

(14)

Substituting from eq.(11b) into eq. (14) gives

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) = -\int_0^L \mu(x)\left[\sum_{l=1}^{N} \omega^2 q_l W_l(x)\right] W_i(x)dx .$$

(15)

Similarly, the potential strain energy is found from

$$y''(x) \equiv \frac{\partial^2 y}{\partial x^2} = \sum_{i=1}^{N} q_i W_i''(x)$$

(16)

Therefore,

$$\frac{\partial U}{\partial q_i} = \frac{1}{2}\frac{\partial}{\partial q_i}\int_0^L EI(x)\left[\sum_{i=1}^{N} q_i W_i''(x)\right]^2 dx, \text{ or}$$

$$= \int_0^L EI(x)\left[\sum_{l=1}^{N} q_l W_l''(x)\right] W_i''(x)dx \qquad (17)$$

Substituting eq's. (15) and (17) into eq. (7) results in

$$\int_0^L \mu(x)\left[\sum_{l=1}^{N} \omega^2 q_l W_l(x)\right] W_i(x)dx = \int_0^L EI(x)\left[\sum_{l=1}^{N} q_l W_l''(x)\right] W_i''(x)dx \qquad (18)$$

Then, there will be N equations of the form

$$\sum_{l=1}^{N} q_l \int_0^L \left[\mu(x)\omega^2 W_l(x)W_i(x) - EI(x)W_l''(x)W_i''(x)\right]dx = 0$$

(19)

Equations (19) can be easily recast in matrix form. Let the integrals be called J_{il} . Then,

$$\left[J_{ilma}\omega^2 - J_{ilst}\right]\{q_l\} = 0, \qquad (20a)$$

where

$$J_{ilma} \equiv \int_0^L \mu(x)W_l(x)W_i(x)dx, \text{ and}$$

$$J_{ilst} \equiv \int_0^L EI(x)W_l''(x)W_i''(x)dx . \qquad (20b)$$

It can be shown[1] that the only nontrival solution to eq. (20a) occurs when ω takes on a value ω_n which causes

$$\det\left[J_{ilma}\omega^2 - J_{ilst}\right]=0$$

(21)

Otherwise, the $\{q_l\}=0$. Then,

$$\left[\left[J_{ilma}\right]^{-1}\left[J_{ilst}\right]- \omega^2 I\right]\{q\}=0$$

(22)

Expanding eq.(22) leads to the characteristic equation for this problem. If three assumed modes are used, it is a cubic polynomial in ω^2. Instead, FLEXIT plots eq.(21) as a function of ω^2. The observed approximate zero crossings are then iteratively refined.

Normal Mode Shapes

The normal mode shapes are derived from eq's. (6) and (19). Suppose that ω took on one of its solution values, say the kth natural frequency. Then, for three assumed modes, a normal mode looks like

$$\eta(x)=W_1(x)+\frac{q_2}{q_1}W_2(x)+\frac{q_3}{q_1}W_3(x).$$

(23)

since all that can be determined from unforced oscillations are the relative amplitudes of the contributions from the various assumed mode shapes. Equation (19) can be expanded to give

$$\left[J_{11ma}\omega^2 - J_{11st}\right]+\frac{q_2}{q_1}\left[J_{12ma}\omega^2 - J_{12st}\right]+\frac{q_3}{q_1}\left[J_{13ma}\omega^2 - J_{13st}\right]=0,$$

$$\left[J_{21ma}\omega^2 - J_{21st}\right]+\frac{q_2}{q_1}\left[J_{22ma}\omega^2 - J_{22st}\right]+\frac{q_3}{q_1}\left[J_{23ma}\omega^2 - J_{23st}\right]=0 \text{ , and}$$

(24)

$$\left[J_{31ma}\omega^2 - J_{31st}\right]+\frac{q_2}{q_1}\left[J_{32ma}\omega^2 - J_{32st}\right]+\frac{q_3}{q_1}\left[J_{33ma}\omega^2 - J_{33st}\right]=0.$$

By implication, ω must take on one of the values ω_n for there to be a solution at all.

Then, any two, say the first two, of the three equations above can be solved for $\frac{q_2}{q_1}$ and $\frac{q_3}{q_1}$:

$$\frac{q_2}{q_1} = \frac{A_{23}A_{11} - A_{21}A_{13}}{A_{22}A_{13} - A_{23}A_{12}} \text{ and } \frac{q_3}{q_1} = \frac{A_{12}A_{21} - A_{11}A_{22}}{A_{22}A_{13} - A_{23}A_{12}} \text{ where } A_{ij} = J_{ijma}\omega^2 - J_{ijst}\,.$$

(25)

The mode shape is governed by the choice of ω_n.

.

FLEXIT Implementation

Nonuniform beam properties are assumed to vary linearly with x across a segment. In general such a beam is composed of segments in which $\mu(x)$ and $EI(x)$ are defined at the segment end points, and, neither $\mu(x)$ nor $EI(x)$ are continuous across segment boundaries. Trapezoidal integration is used to evaluate the integrals in eq. (20b). The desired granularity (number of integration steps InS along the body) is input. The number of steps used for an individual segment is based on

INT[$\frac{L_{SEG}}{L_{BODY}} InS$]+1. This ensures that at least the desired number of steps are used.

The normal modes, eq.(5) and their second derivatives, are generated by external functions:

$$+\left(\frac{\omega_n}{a}\right)\left[\cosh(\sqrt{\omega_n/a}\,x) - \cos(\sqrt{\omega_n/a}\,x)\right]$$

(26)

Note that ω_n appearing in eq's. (5) and (26) above are the natural frequencies for a uniform beam as given by eq. (4b).

It is customary to normalize the natural mode shapes. There are many ways to do this. In FLEXIT they are normalized to have the same nose tip value (2) as the uniform beam modes.

FLEXIT uses a simple bisection algorithm to find the roots of the characteristic equation. This requires the user to make a rough guess on these roots, and then set upper and lower bounds for the bisection iteration. There are two ways to do this. First, use the roots for an roughly equivalent uniform beam as a guide. The second is to plot the determinant, eq. (21), vs. the square of the frequency. The zero crossings of this plot are the roots being sought. The following table compares the uniform beam results for the corresponding nonuniform case. The rocket analyzed is the Space Vector Aries, Minuteman II second stage converted into a sounding rocket. As can be seen, the uniform beam results can be useful to guide the search for roots to the characteristic equation.

Mode	Fundamental natural frequency, rad/sec	First Harmonic natural frequency, rad/sec	Second Harmonic natural frequency, rad/sec
Nonuniform	257.632	585.471	1523.704

Beam			
Uniform Beam	277.452	764.806	1499.317

Since our major practical interest is in the fundamental mode, only the first three uniform beam modes are carried in the calculation, and only the first two mode shapes are estimated and plotted.

References

1. F. B. Hildebrand, "Methods of Applied Mathematics", Prentice-Hall, Inc.,Englewood Cliffs, N. J., 1956.
2. R.L. Bisplinghoff, H. Ashley, and R. L. Halfman, "Aeroelasticity", Addison-Wesley Publishing Company, Cambridge, 1955